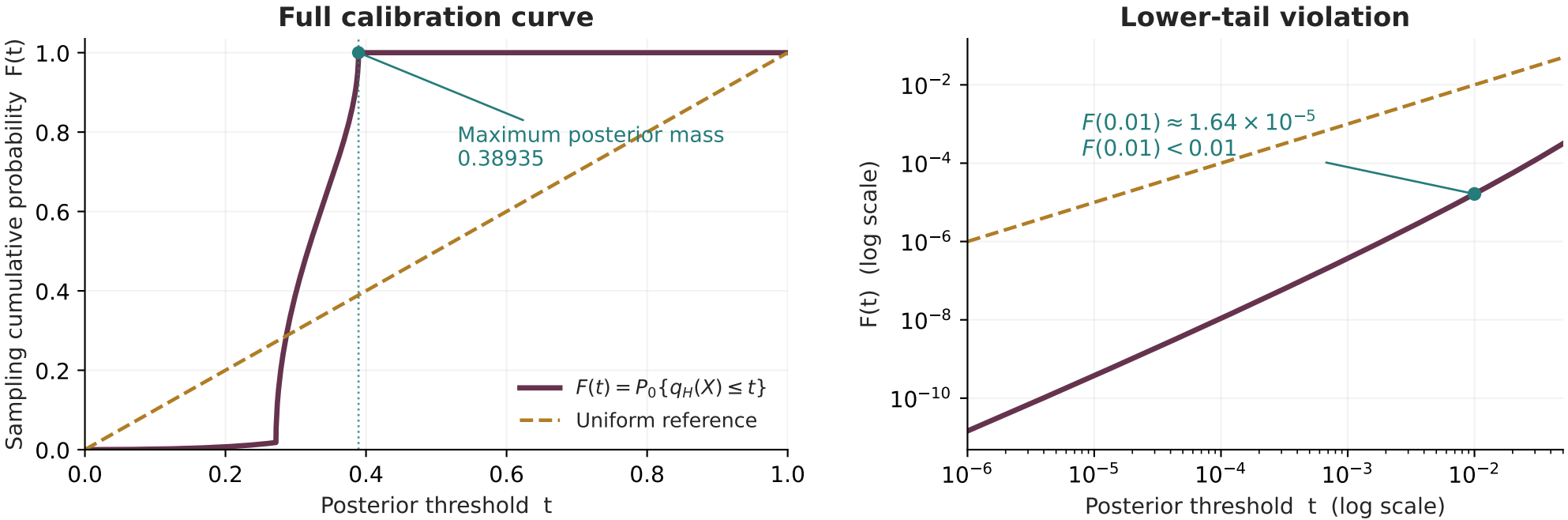
One fixed false hypothesis can cross the uniform calibration line



$H = (-2, -1) \cup (1, 2)$; $X \sim N(0, 1)$; $Q_x = N(x, 1)$. Normal-CDF roots; no simulation.